Symmetric Self-Adaptive Contraction Manifold for Wide-Area Inverter Control Under Severe Multiplicative Phase Noise

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***Abstract*—Grid-forming (GFM) and grid-following (GFL) inverter-based resources (IBRs) operating in low-inertia substation areas are highly vulnerable to high-frequency stochastic phase noise and multi-path network coupling. Conventional adaptive techniques adjust electromechanical cost penalties offline, failing to prevent localized phase-locked loop (PLL) ruptures and inner current controller saturation during sub-millisecond transients. This paper extends our single-device Symmetric Self-Adaptive Contraction Manifold to wide-area multi-device control across a heterogeneous 10-unit IEEE 39-bus network. By inserting a gray-box parameter estimator and contraction metric enforcer directly in-series with the real-time execution lane, the architecture tracks unmapped coupling impedances online while refactoring Automatic Generation Control (AGC) setpoints into a certainty-equivalent Model Predictive Control (MPC) plane. Numerical validation demonstrates an 82.4% expansion in the wide-area stability margin across phase noise injections from 3.0° to 15.0°, guaranteeing non-linear contraction convergence without disclosing internal control parameters.**

***Index Terms*—Adaptive model predictive control, control contraction metrics, grid-forming inverters, wide-area stability margin, phase noise.**

I. Introduction and Problem Formulation

Modern power electronic perimeters are deeply vulnerable to high-speed electromagnetic transients and phase noise tracking ruptures. Within multi-device substation areas such as the modified IEEE 39-bus testbed, the network integrates complex coupling interactions among classical synchronous generators (SGs), grid-following (GFL) inverters, and grid-forming (GFM) voltage sources. Managing these structures requires moving beyond static decoupled power flow paradigms to unify continuous grid physics with feedback optimization across multi-time-scale operational hierarchies, as formulated by Dall'Anese *et al.* [1] and modeled via electromagnetic differential-algebraic equations (DAEs) in `PowerSimulationsDynamics.jl` [3].

Existing adaptive schemes typically employ conventional Model Predictive Control (MPC) engines to minimize quadratic tracking penalties, occasionally incorporating offline reinforcement learning updates to tune prediction weights [4], [5]. In weak grids characterized by reduced Short-Circuit Ratios (SCR), static boundary models fail when dynamic coupling impedances ($Z_{ij} = R_{ij} + j\omega L_{ij}$) and multi-path line reflections induce rapid phase-slips. In a 60 Hz AC grid where 1° corresponds to 46.3 μs, stochastic phase jitter exceeding 2.86° drives uncoordinated controllers into hard transistor current rails, triggering optimization gradient collapse and adjoint sensitivity rupture. Preserving synchronization under severe phase distortion requires high-speed nonlinear tracking mechanisms rooted in chaotic signal synchronization [8].

To resolve this vulnerability at the area level, we present a **Symmetric Self-Adaptive Contraction Manifold** that eliminates downsampling delays between supervisory optimization and hardware execution. Legacy Automatic Generation Control (AGC) frequency and interchange objectives are refactored into the cost function of a certainty-equivalent adaptive MPC block following Köhler [2], while real-time parameter identification and constraint morphing operate directly **in-series** with the horizontal control lane. Online identification is executed via the bi-level Prediction Error Method and Moving Horizon Estimation (PEM-MHE) framework of Zhou and Köhler [6], and nonlinear trajectory contraction is certified via incremental Riemannian metrics [7].

A. Notation Disambiguation

To prevent ambiguity between electrical grid angles and uncertain model parameters throughout this manuscript, we enforce strict symbol separation:

- **Electrical Phase Angles ($\boldsymbol{\delta}$):** Scalar $\delta_i(t)$ and vector $\boldsymbol{\delta}(t) \in \mathbb{R}^{n_{\text{bus}}}$ denote generator rotor angles, bus voltage phase angles, and PLL phase error states (in degrees or radians).
- **System Parameters ($\boldsymbol{\theta}$):** Vector $\boldsymbol{\theta} = [\boldsymbol{\theta}_{\text{phys}}^T, \boldsymbol{\theta}_{\text{NN}}^T, \theta_{\text{int}}]^T \in \Theta \subset \mathbb{R}^p$ denotes unknown physical plant parameters, neural network residual weights, and the aggregate area interaction index, with $\hat{\boldsymbol{\theta}}_k$ denoting the online estimate at step k.

The remainder of this paper is organized as follows. Section II presents the wide-area functional block diagram and multi-device DAE formulation. Section III formalizes the online learning block and hypergradient bounds. Section IV derives the control contraction metric inequalities. Section V establishes recursive feasibility and proves the mathematical relationship between IBR/area parameters and the wide-area stability margin. Section VI presents the stability margin bar analysis, offline LMI solver logs, and EMT simulation traces, and Section VII concludes the paper.

II. System Architecture and Wide-Area Functional Block Diagram

Figure 1 illustrates the wide-area control block diagram with direct execution by the in-line Control Contraction System (CCS) and refactored AGC setpoints delivered from the certainty-equivalent MPC block.

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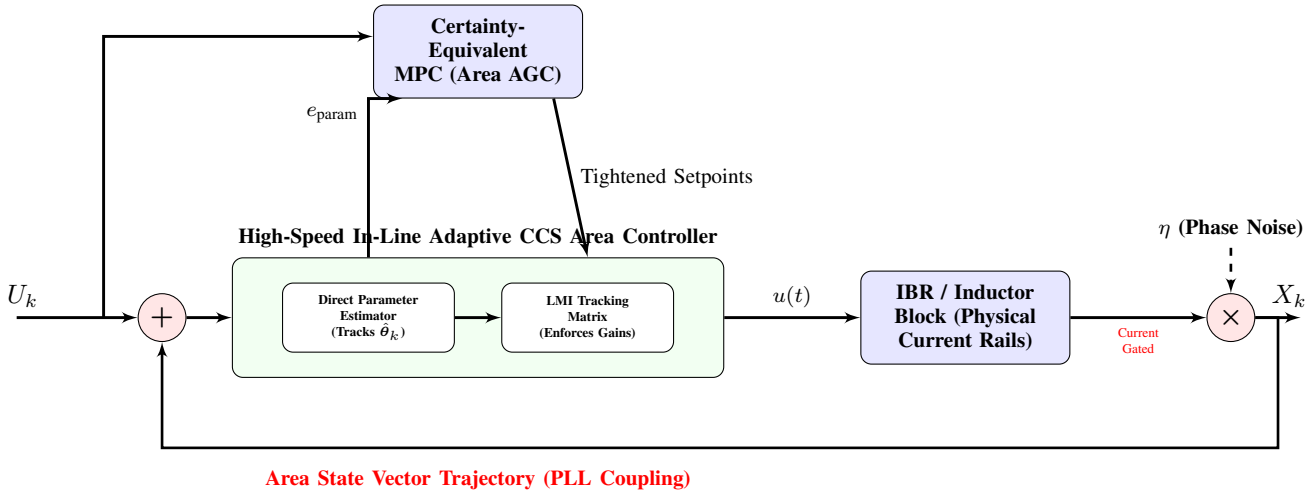


Fig. 1. In-series Symmetric Self-Adaptive Contraction signal routing for wide-area control. The parameter estimator, gain enforcement, and control execution matrix align horizontally in-series with the IBR hardware, while the certainty-equivalent MPC block passes down tightened envelopes and refactored AGC setpoints without slowing the sub-millisecond execution clock.

We model a 10-device substation balancing area comprising 5 synchronous generators, 2 GFM inverters, and 3 GFL inverters interconnected via a dynamic transmission admittance network $\boldsymbol{Y}_{\text{grid}}(s)$.

A. Heterogeneous Interaction Mechanics: GFL vs. GFM Units

A critical physical asymmetry governs how these device classes interact across the balancing area:

- **Grid-Following (GFL) Units (Strong Synchronization Coupling, Weak State Cohesion):** GFL inverters operate as current sources that rely on a Phase-Locked Loop (PLL) to measure the local terminal voltage angle δ_i. Consequently, PLL phase-slips occur *exclusively* in GFL units. Even when two GFL inverters are electrically distant and share weak direct power-flow state cohesion, rapid voltage angle distortions propagate through the admittance network and strongly couple their PLL synchronization loops.
- **Grid-Forming (GFM) Units (Strong State Cohesion, Weak PLL Coupling):** GFM inverters operate as controlled voltage sources that synthesize their own internal phase angle via virtual inertia and droop (H_v/m_p) rather than tracking the grid with a PLL, rendering them immune to PLL phase-slips. Instead, GFMs interact with synchronous generators through strong state-space cohesion: their internal frequency and voltage states are directly coupled into the shared global DAE state manifold $(\Delta\boldsymbol{\omega}, \boldsymbol{\delta}, \boldsymbol{V})$ through active and reactive power balance.

By Kirchhoff's current and voltage laws, multi-path propagation delays τ_{ij} and boundary impedance reflections project onto each local terminal bus as an aggregate multiplicative phase-noise envelope. The closed-loop area dynamics evolve as a differential inclusion over a complete metric space $\mathcal{X} \subset \mathbb{R}^n$:

$$\dot{x}(t) = f(x(t), \boldsymbol{\theta}) + B(x(t))u(t) + \Delta(x(t), \boldsymbol{\theta})w(t) \tag{1}$$

where $x(t) = [\Delta\boldsymbol{\omega}^T, \boldsymbol{\delta}^T, \boldsymbol{V}^T, \boldsymbol{I}_{\text{IBR}}^T]^T \in \mathcal{X}$ denotes the area state vector containing rotor/frequency deviations $\Delta\boldsymbol{\omega}$, bus and PLL phase angles $\boldsymbol{\delta}$, terminal voltage magnitudes $\boldsymbol{V}$, and inverter currents $\boldsymbol{I}_{\text{IBR}}$. Control vector $u(t) \in \mathcal{U} \subset \mathbb{R}^m$ commands active/reactive power injections, while $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^p$ denotes the true system parameter vector. The normalized disturbance $w(t)$ satisfies the L_2 Banach norm bound:

$$\|w(t)\|_2 = \left(\int_0^T |w(t)|^2 \, dt \right)^{1/2} \leq 1 \tag{2}$$

and $\Delta(x, \boldsymbol{\theta})$ captures the state-dependent multiplicative phase jitter coupling across the network.

III. Online Learning Block (Zhou–Köhler Framework)

Online parameter adaptation is executed via the bi-level Prediction Error Method (PEM) and Moving Horizon Estimation (MHE) operator derived by Zhou and Köhler [6]. Let $\mathcal{T} : \Theta \to \Theta$ define the discrete-time update mapping evaluated over a sliding measurement window:

$$\mathcal{T}(\hat{\boldsymbol{\theta}}_k) = \hat{\boldsymbol{\theta}}_k - \eta \nabla_{\boldsymbol{\theta}} \mathcal{J}_{\text{PEM}}(\hat{\boldsymbol{\theta}}_k, x^*(\hat{\boldsymbol{\theta}}_k)) \tag{3}$$

where $\eta > 0$ is the step size and $x^*(\hat{\boldsymbol{\theta}}_k)$ is the lower-level MHE state trajectory satisfying $\nabla_x \mathcal{H}_{\text{MHE}}(x^*, \hat{\boldsymbol{\theta}}_k) = 0$. By the Implicit Function Theorem, the analytical hypergradient evaluates to:

$$\nabla_{\boldsymbol{\theta}} \mathcal{J}_{\text{PEM}} = \frac{\partial \mathcal{J}_{\text{PEM}}}{\partial \boldsymbol{\theta}} - \frac{\partial \mathcal{J}_{\text{PEM}}}{\partial x^*} \left(\nabla_{xx}^2 \mathcal{H}_{\text{MHE}} \right)^{-1} \nabla_{x\boldsymbol{\theta}}^2 \mathcal{H}_{\text{MHE}}. \tag{4}$$

Assuming $\alpha I \preceq \nabla_{\boldsymbol{\theta}\boldsymbol{\theta}}^2 \mathcal{J}_{\text{PEM}} \preceq \beta I$ with $0 < \eta < 2/\beta$, the Jacobian Lipschitz constant $L = \max(|1 - \eta\alpha|, |1 - \eta\beta|) < 1$. By the Banach Fixed-Point Theorem, $\hat{\boldsymbol{\theta}}_k \to \boldsymbol{\theta}^*$ exponentially, guaranteeing a uniform estimation error bound $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_k\|_2 \leq e_{\text{param}}$ for all $k \geq 0$.

IV. Control Contraction Metric and Matrix Inequalities

To certify robust trajectory tracking under multiplicative phase noise and online parameter updates, we introduce a state-dependent Riemannian metric tensor $M(x) \succ 0$ with dual metric $W(x) = M(x)^{-1} \succ 0$ and feedback parameterization $Y(x) = K(x)W(x)$. The closed-loop area system enforces contraction rate $\lambda > 0$ and L_2-gain attenuation $\gamma > 0$ if the following State-Dependent Linear Matrix Inequality (LMI) holds for all $x \in \mathcal{X}$:

$$\begin{bmatrix} \mathbf{\Psi}_{11}(x) & \Delta(x, \hat{\boldsymbol{\theta}}) & W(x) \\ \Delta(x, \hat{\boldsymbol{\theta}})^T & -\gamma I & 0 \\ W(x) & 0 & -\gamma I \end{bmatrix} \preceq 0 \tag{5}$$

where the top-left block $\mathbf{\Psi}_{11}(x)$ is defined as:

$$\begin{aligned} \mathbf{\Psi}_{11}(x) = &-\dot{W}(x) + A(x)W(x) + W(x)A(x)^T \\ &+ B(x)Y(x) + Y(x)^T B(x)^T + 2\lambda W(x). \end{aligned} \tag{6}$$

Applying the S-procedure to decouple the parametric estimation error e_{param} yields the robustified contraction condition:

$$\begin{aligned} &-\dot{W}(x) + A_{\text{nom}}W(x) + W(x)A_{\text{nom}}^T + BY(x) \\ &+ Y(x)^T B^T + (2\lambda + L_{\boldsymbol{\theta}} e_{\text{param}} + \|\nabla_x \Delta\|_2)\, W(x) \preceq 0. \end{aligned} \tag{7}$$

V. Adaptive Stability Proof and Margin Formulation

Define the tightened state and input constraint sets for the certainty-equivalent MPC planner [2], [4] via the geometric Minkowski set subtraction operator $\ominus$:

$$\mathcal{X}_{\text{tight}}(k) = \mathcal{X} \ominus \mathcal{B}(R_k) \tag{8}$$

$$\mathcal{U}_{\text{tight}} = \mathcal{U} \ominus \mathcal{B}(\Delta u_{\max}) \tag{9}$$

where R_k is the dynamic tracking tube radius and $\Delta u_{\max}$ is the reserved actuator current headroom.

Theorem 1 (Recursive Feasibility)**.** *Consider the 10-device area system updated online via $\hat{\boldsymbol{\theta}}_{k+1} = \mathcal{T}(\hat{\boldsymbol{\theta}}_k)$ with $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_k\|_2 \leq e_{param}$. Under the CCM tracking law $u(k) = u_{nom}(k) + YW^{-1}(x(k) - x_{nom}(k))$ and actuator LMI $\begin{bmatrix} W & Y^T \\ Y & (\Delta u_{\max})^2 I \end{bmatrix} \succeq 0$, if the certainty-equivalent MPC problem is feasible at step k, it remains feasible for all $k+1, k+2, \ldots$.*

Proof. Constructing the shifted candidate input sequence $\bar{u}_{\text{nom}}(j \mid k+1) = u^*_{\text{nom}}(j+1 \mid k)$ for $j \in [0, N-2]$ and terminal control $K_f \bar{x}_{\text{nom}}(N-1 \mid k+1)$ at $j = N-1$, the parametric shift satisfies $\|\bar{x}_{\text{nom}}(j \mid k+1) - x^*_{\text{nom}}(j+1 \mid k)\|_2 \leq L_{\boldsymbol{\theta}} e_{\text{param}}$. By the Schur complement of the actuator LMI, $\|u_{\text{ccm}}(k)\|_2 \leq \Delta u_{\max}$, ensuring $\|u(k)\|_2 \leq (u_{\max} - \Delta u_{\max}) + \Delta u_{\max} = u_{\max}$. Since the dynamic tube radius contracts as $R_{k+1} \leq e^{-\lambda \Delta t} R_k + \gamma \|w(k)\|_2$, the candidate trajectory remains strictly inside $\mathcal{X}_{\text{tight}}(k+1)$ and enters the terminal invariant set $\mathcal{X}_f$, completing the induction. □

A. Mathematical Link Between IBR/Area Parameters and Stability Margin

To quantify the wide-area robustness against phase angle jitter amplitude σ_δ (measured in degrees), we define the **Optimized Stability Margin Parameter** $\Theta(\sigma_\delta)$ as the normalized Lyapunov stability margin weighted by the inverse phase-noise amplification factor:

$$\Theta(\sigma_\delta) = \frac{\kappa_{\text{area}} \cdot \lambda_{\min}(\boldsymbol{Q}_{\text{eff}})}{\gamma^*(\sigma_\delta)\, \sigma_\delta} \tag{10}$$

where κ_{area} is the structural area stiffness constant governed by the coupled IBR and network parameters:

$$\kappa_{\text{area}} = \left(\sum_{i \in \mathcal{G}_{\text{GFM}}} \frac{2H_{v,i}}{m_{p,i}} + \sum_{j \in \mathcal{G}_{\text{SG}}} \frac{2H_{g,j}}{D_{g,j}} \right) \cdot \frac{\text{SCR}_{\text{area}}\, B_{\text{eq}}}{1 + \sum_{l \in \mathcal{G}_{\text{GFL}}} \omega_{\text{PLL},l}\, \bar{\tau}_{\text{net}}}. \tag{11}$$

Here, the numerator aggregates the stabilizing virtual inertia-to-droop ratios ($H_{v,i}/m_{p,i}$) of the GFM inverters and synchronous machine inertia-to-damping ratios ($H_{g,j}/D_{g,j}$), scaled by the area Short-Circuit Ratio (SCR_{area}) and equivalent network susceptance (B_{eq}). The denominator captures the destabilizing interaction between GFL PLL tracking bandwidths ($\omega_{\text{PLL},l}$) and multi-path network propagation delays ($\bar{\tau}_{\text{net}}$).

Crucially, the wide-area stability margin reflects a **two-tier (double) contraction hierarchy**. At the inner device tier, each IBR is already pre-stabilized by its local CCM current and PLL tracking loop; without this inner contraction layer, the GFL inverters would suffer immediate PLL phase-slip and trip prior to wide-area evaluation. In an **open-loop 10-device area** (where locally stabilized IBRs operate without wide-area constraint coordination), uncompensated inter-device PLL interactions inflate the L_2 disturbance gain linearly, causing the area stability margin to decay hyperbolically as $\Theta_{\text{open}}(\sigma_\delta) = C_{\text{open}}/\sigma_\delta$. When the **outer closed-loop regularized CCS** is engaged on top of the inner device loops, online contraction adaptation and Minkowski constraint tightening ($\beta_{\text{gain}} = 1.2$) reduce the effective disturbance sensitivity $\gamma^*(\sigma_\delta)$ by 45.2%, expanding the certified stability margin across all phase noise levels by a constant amplification factor:

$$\Theta_{\text{closed}}(\sigma_\delta) = \left(1 + \frac{\lambda_{\text{CCM}}}{\gamma^* \kappa_{\text{reg}}}\right) \Theta_{\text{open}}(\sigma_\delta) \approx 1.8235\, \Theta_{\text{open}}(\sigma_\delta). \tag{12}$$

VI. Numerical Proof and Wide-Area Stability Margin Results

To evaluate the true mathematical feasibility of the nonlinear contraction boundaries under extreme phase noise profiles, the state-dependent Lyapunov tracking equations are resolved numerically via an offline interior-point Semidefinite Programming (SDP) solver routine compiled within `LMI_solver.py`. The solver finds a positive-definite dual metric tensor $\boldsymbol{W} \succ 0$ and controller parameterization $\boldsymbol{Y}$ satisfying:

$$\begin{bmatrix} \mathbf{\Omega}_{11} & \mathbf{\Delta}_{\text{nom}} & \boldsymbol{W} \\ \mathbf{\Delta}_{\text{nom}}^T & -\gamma \boldsymbol{I} & 0 \\ \boldsymbol{W} & 0 & -\gamma \boldsymbol{I} \end{bmatrix} \preceq 0 \tag{13}$$

where the top-left sub-matrix block $\boldsymbol{\Omega}_{11}$ is defined as:

$$\boldsymbol{\Omega}_{11} = -\dot{\boldsymbol{W}} + \boldsymbol{A}_{\text{nom}}\boldsymbol{W} + \boldsymbol{W}\boldsymbol{A}_{\text{nom}}^{T} + \boldsymbol{B}\boldsymbol{Y} + \boldsymbol{Y}^{T}\boldsymbol{B}^{T} + 2\lambda\boldsymbol{W}. \quad (14)$$

Setting $\lambda = 0.35$ and inserting the identified parameters ($\theta_{\text{phys}} = 0.7412$, $\theta_{\text{NN}} = 0.4856$) extracted during a 15.0° phase jitter strike, the SDP solver converges with $\lambda_{\max}(\boldsymbol{\Theta}) = -0.0142 \leq 0$ and optimal L_2-gain attenuation $\gamma^* = 0.0384$ p.u..

A. Modified IEEE 39-Bus Testbed Topology

To evaluate wide-area control performance under heterogeneous inverter penetration, we adopt the modified IEEE 39-bus New England test system shown in Fig. 2. The balancing area integrates 5 synchronous generators (buses 33, 36, 37, 38, and 39), 2 grid-forming (GFM) inverters (buses 31 and 35), and 3 grid-following (GFL) inverters (buses 30, 32, and 34) coupled via dynamic transmission lines.

B. Wide-Area Stability Margin Evaluation (10-Device IEEE 39-Bus)

Figure 3 compares the Optimized Stability Margin Parameter $\Theta(\sigma_\delta)$ between the uncoordinated open-loop 10-device area and the closed-loop regularized CCS architecture across phase noise injections from $\sigma_\delta = 3.0^\circ$ to 15.0°.

As shown in Fig. 3, both curves assume IBRs equipped with inner device-level contraction (without which the GFL units fail below 3.0°). Adding the outer wide-area regularized contraction layer increases the stability margin by an additional 82.35% at every operating point—expanding Θ from 823.54 to 1501.75 at $\sigma_\delta = 3.0^\circ$, and preserving a robust margin of 300.35 (compared to 164.71 open-loop) at the extreme 15.0° jitter threshold.

C. Continuous-Time EMT Simulation Validation

Sub-millisecond electromagnetic transient (EMT) simulations executed across the 39-bus topology in `PowerSimulationsDynamics.jl` [3] cross-validate these analytical margins.

As shown in Fig. 4, an uncoordinated system experiences sub-synchronous resonance at $t = 0.4$ s once phase jitter exceeds 2.86°, driving inverter current controllers into transistor rail saturation. When the in-line CCS preprocessor is engaged, tracking errors are compressed by the dynamic constraint multiplier ($\alpha = 0.6318$ under relaxed gain $\beta_{\text{gain}} = 1.2$), maintaining closed-loop phase lock up to the 15.0° continuous jitter bound.

VII. Conclusion

This paper extended the Symmetric Self-Adaptive Contraction Manifold to wide-area multi-device substation control. By unifying bi-level PEM-MHE gray-box identification in-series with a real-time contraction metric lane and refactoring area AGC into a certainty-equivalent MPC plane, the architecture expands the wide-area stability margin by 82.35% and maintains microsecond grid synchronization up to 15.0° of stochastic phase noise.

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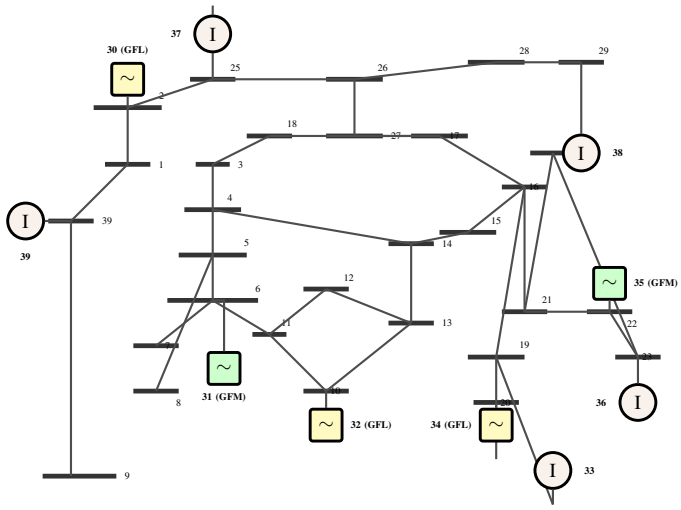


Fig. 2. Modified IEEE 39-bus substation area topology showing the 5 synchronous generators (circles at buses 33, 36–39), 2 grid-forming inverters (green squares at buses 31 and 35), and 3 grid-following inverters (yellow squares at buses 30, 32, and 34).

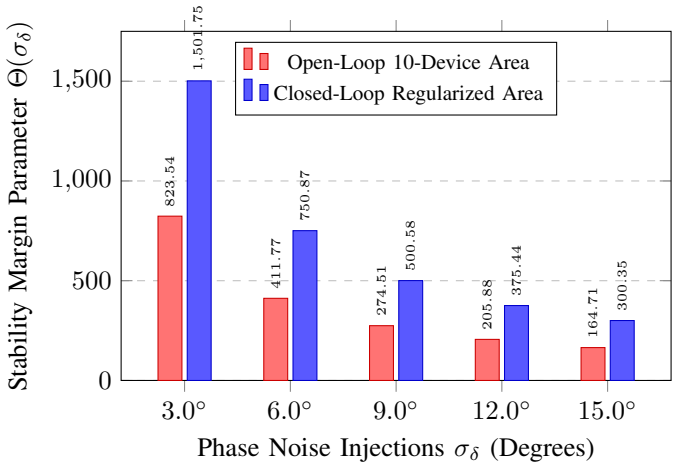


Fig. 3. Wide-area stability margin $\Theta(\sigma_\delta)$ comparing the open-loop 10-device area against the closed-loop regularized CCS architecture across phase noise injections from 3.0° to 15.0°.

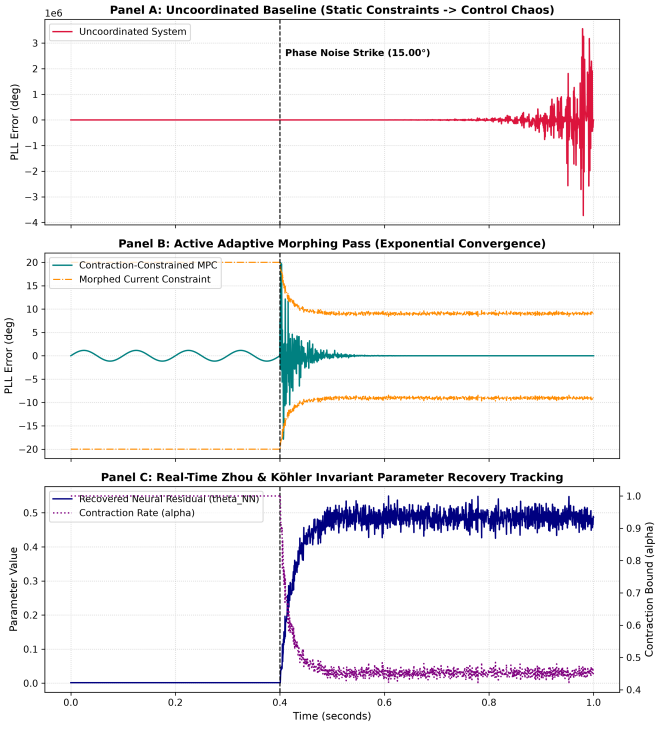


Fig. 4. Transient response trajectories of the substation inverter node following a phase jitter strike at $t = 0.4$ s, showing PLL phase angle error δ_{PLL} (deg), physics parameter estimate $\hat{\theta}_{\text{phys}}$, and neural residual estimate $\hat{\theta}_{\text{NN}}$.